

Lecture Notes for Neural Networks I



Visualization II

Class Logistics and Agenda

- Logistics:
 - Lab One!
- Agenda:
 - Finish Visualization
 - Town Hall
- Next Time: **Flipped Module One!**
 - Click on the link in canvas, Main Page
 - **Individually:** Watch video, Take Quiz/Notes
 - Come ready to download, work on, and turn in a Jupyter notebook on the topics
 - We will work through the first questions as a class

ICA1 - Linear Regression

$J(w) = \frac{1}{2} \sum_i (y^{(i)} - w^T x^{(i)})^2$

$\vec{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix}, X = \begin{bmatrix} \leftarrow x^{(1)T} \rightarrow \\ \leftarrow x^{(2)T} \rightarrow \\ \vdots \\ \leftarrow x^{(n)T} \rightarrow \end{bmatrix}$

$J(w) = \frac{1}{2} (Xw - y)^T (Xw - y)$

$\nabla_w J(w) = \frac{1}{2} X^T (Xw - y)$

$X^T X w = X^T y$

And that means that even in the b

$N = (X^T X)$

$Xw = \begin{bmatrix} x^{(1)T} w \\ x^{(2)T} w \\ \vdots \\ x^{(n)T} w \end{bmatrix}$

ASIDE

$0 = \frac{d}{dw} (Xw - y)^T (Xw - y)$

$= 2 X^T (Xw - y)$

2. Module Two: Linear Classification (Click for Resources)

Flipped Module: Linear Regression

Logistic Regression I

Lecture Video Supplement

Class Overview, by topic

Table Data
Visualization

Numpy, Pandas, Seaborn
Overviews with some in-depth discussion

Linear and
Logistic
Regression

Numpy, Recreate API for Scikit-learn
Detailed mathematics for simple optimization
intuition for advanced optimization

Neural Networks
and Back Prop.

Numpy, Recreate API for DL Frameworks
Detailed mathematics for NN operations

Wide and Deep
Networks

Convolutional
Networks

Sequential
Networks

Keras, Tensorflow, Torch
Intuition, Detailed implement.

Large Language
Models (if time)

Hugging Face
Intuition only

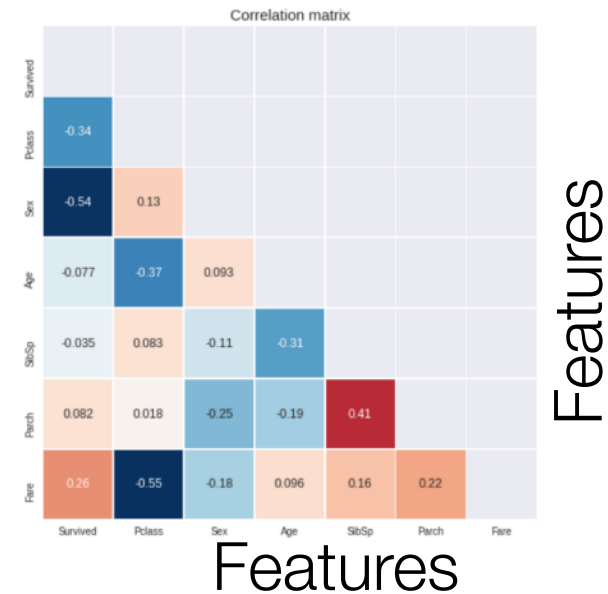
Self Test: What is the MAIN difference in these plots?

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	2	3	male	Q	S	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	0	1	1	0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C	0	0	0	0	0
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S	0	1	0	0	1

- A. These correlation plots are not different
- B. Correlation taken is between rows (left) or columns (right)
- C. The colormap in each is different
- D. The left plot clusters, the right does not.



$$\rho_{x,y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \cdot \sqrt{\sum_i (y_i - \bar{y})^2}}$$



- Focus on conclusions from each graphic
 - Histogram
 - KDE
 - HeatMaps and Correlation
 - Scatter and Scatter Matrix
 - Box / Violin / Swarm



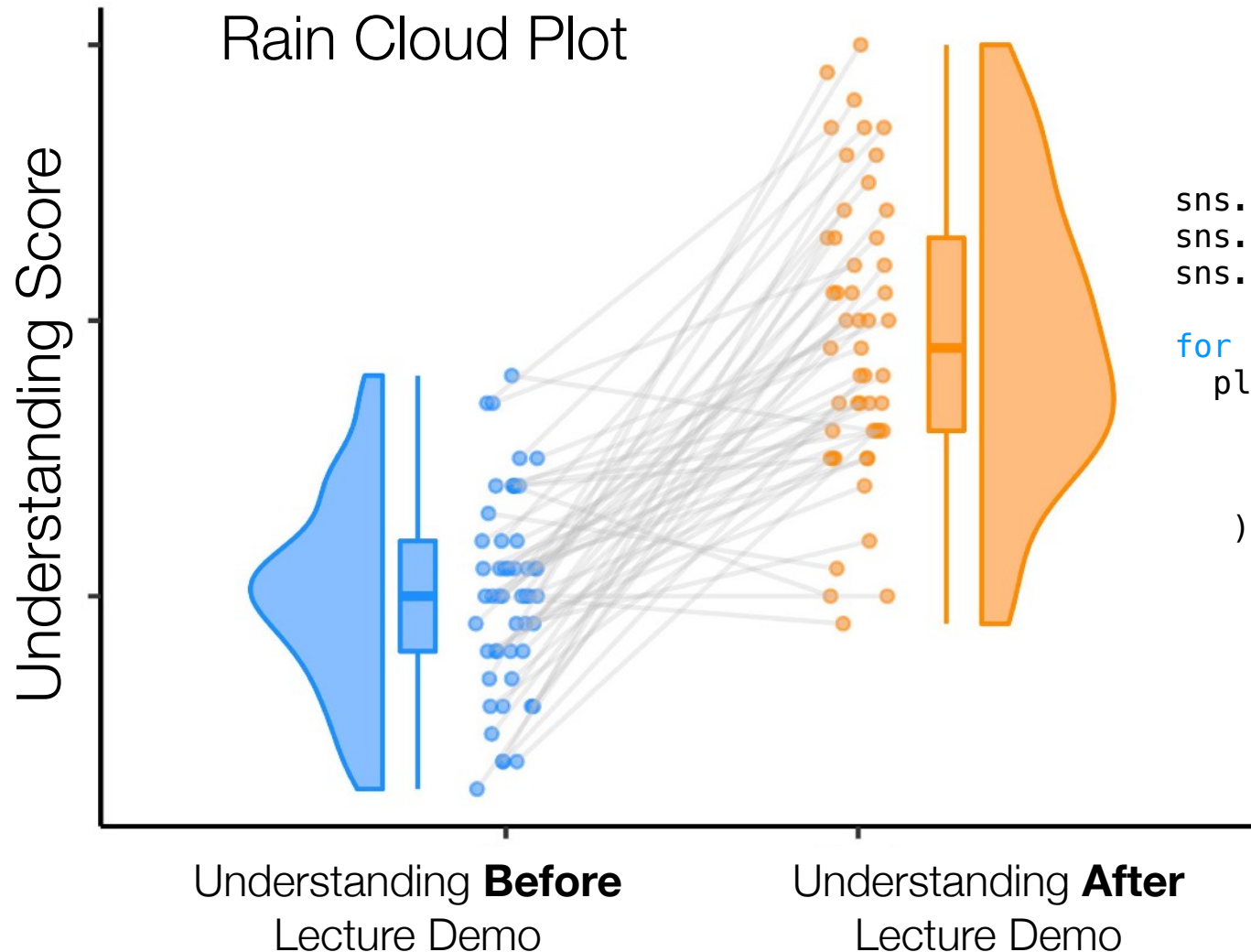
02.Data Visualization.ipynb

Matplotlib

Seaborn

Plotly (if time)

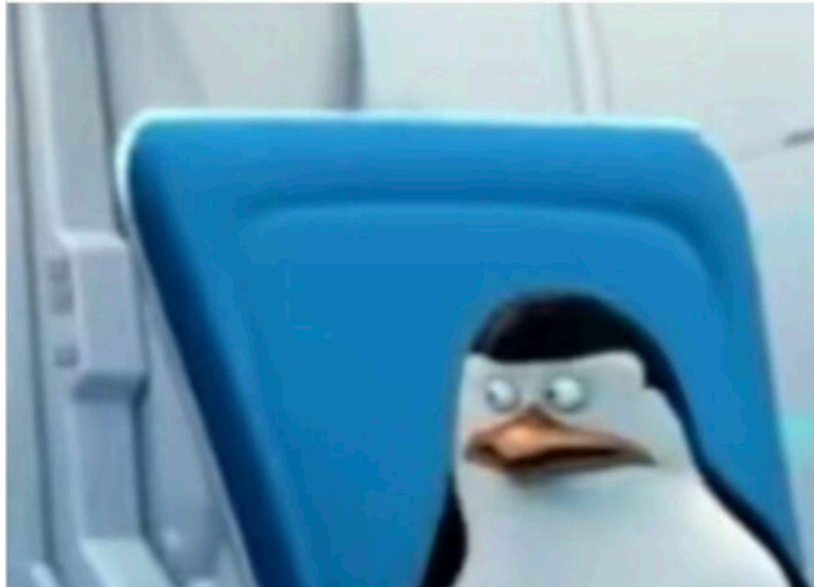
Now you have visualization building blocks



```
sns.violinplot( ... )  
sns.boxplot( ... )  
sns.swarmplot( ... )  
  
for i in range(n_samples):  
    plt.plot(  
        [0, 1],  
        [pre[i], post[i]],  
        color='gray', ...  
    )
```

Lab One: Town Hall

Me looking back making sure no one saw me type $5+3$ on the calculator:



For Next Lecture

- Next Time:
 - Flipped Module One!

Supplemental Slides



Minkowski Distance

- Minkowski Distance is a generalization of Euclidean Distance

$$\mathbf{dist} = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{\frac{1}{r}}$$

Where r is a parameter, n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k th attributes (components) or data objects p and q .

Minkowski Distance: Examples

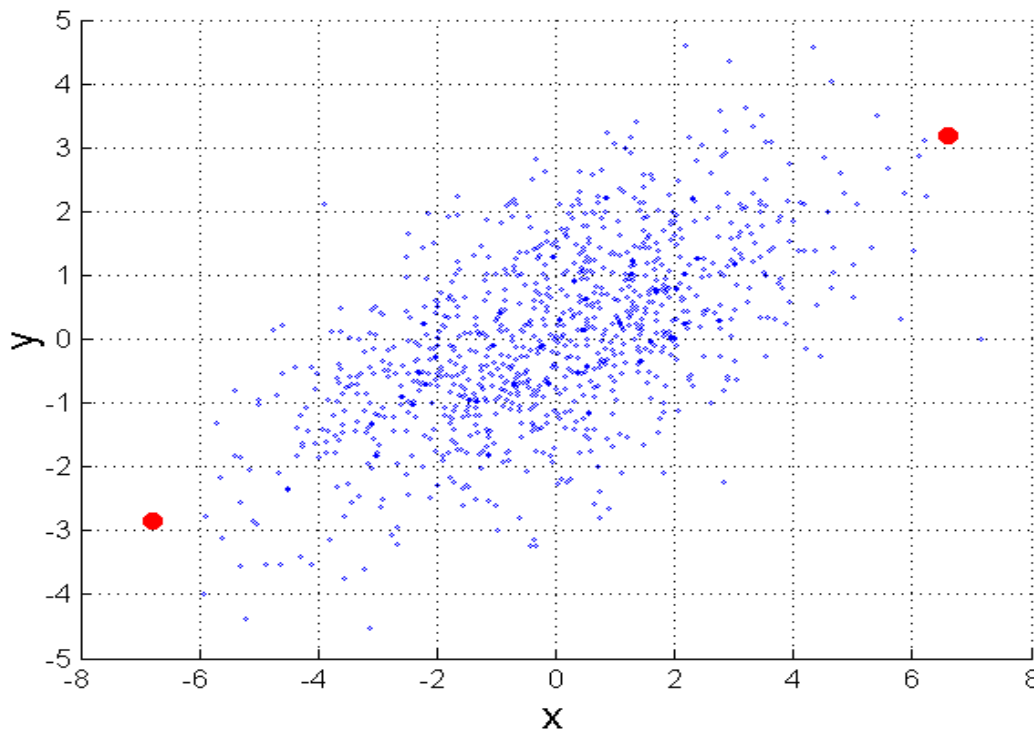
- $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.
 - A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors
- $r = 2$. Euclidean distance (L_2 norm)
- $r \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance.
 - This is the maximum difference between any component of the vectors
- Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

Mahalanobis Distance

$$\text{mahalanobis}(p, q) = (p - q) \Sigma^{-1} (p - q)^T$$

Σ is the covariance matrix of the input data X

$$\Sigma_{j,k} = \frac{1}{n-1} \sum_{i=1}^n (X_{ij} - \bar{X}_j)(X_{ik} - \bar{X}_k)$$



For red points, the Euclidean distance is 14.7, Mahalanobis distance is 6.

Visualization Techniques: Distributions

- Histogram
 - Usually shows the distribution of values of a single variable
 - Divide the values into bins and show a bar plot of the number of objects in each bin.
- KDE
 - Add up Gaussian underneath each point value
 - STD of gaussian is equivalent to number of bins in histogram

